Human-centred intelligent systems and soft computing

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This paper considers the abstract features of human/machine interaction systems that are required for the production of intelligent behaviour. A conceptual architecture is proposed for a subset of intelligent systems called human-centred intelligent systems (HCISs) and it is argued that such systems must be autonomous, robust and adaptive in order to be intelligent. Soft computing is proposed as a promising new technique that can be used to build HCISs, and examples are presented where this is already being done. Finally, flexibility is defined to be a combination of the often-conflicting requirements of robustness and adaptability, and it is argued that the right balance between these two features is necessary to achieve intelligent behaviour.

1. Introduction

Many attempts have been made in the last 40 years to build so-called intelligent systems. An immediate problem is that there is no generally accepted definition of the word intelligent, resulting in long-running debates between AI researchers, philosophers, computer scientists, psychologists, neuroscientists, etc. According to McCarthy, one of the founding fathers of AI, intelligence is the computational part of the ability of an agent to achieve goals in the world [1]. According to Krogman [2], an intelligent system must have a structure to be able to organise itself to learn and adapt to changing situations, as opposed to being simply a program that only reacts in pre-specified situations.

The broad nature of the debate does not lend itself to characterising directly what constitutes intelligent behaviour in specific engineered systems. In this paper, we therefore do not attempt to define the nature of intelligence, but rather adopt an engineering perspective on the issue, similar to that exemplified within AI by Rodney Brooks. More specifically, we define some common abstract attributes or features among systems known to be intelligent (e.g. humans) and argue that artificial systems with such attributes to varying degrees will be perceived by humans to be intelligent to varying degrees. Using an engineering methodology, we identify a number of key features that we consider to be central to intelligence, and then consider how suitable algorithms and techniques that enable such features to be exhibited by computational systems can be developed.

Some of the most important features of intelligent systems, as argued in Azvine et al [3], are autonomy, robustness, and adaptability (see Fig 1).

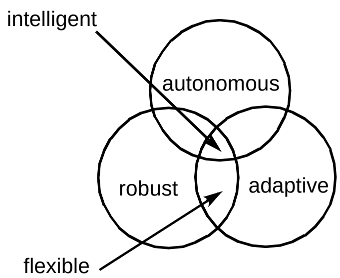

Fig 1 Essential features of an intelligent machine.

Autonomy is needed so that the system can rely on its own experience rather than follow a predefined set of operations. In a general sense, it can mean not being directly under the influence of other systems (or humans). More specifically, autonomy refers to the ability of a system to use perceptual information in conjunction with its built-in knowledge to make decisions in response to changes in its environment in order to meet its goals. The degree of autonomy of a system is a feature that can be improved with time, e.g. a system can rely on its built-in knowledge until it has gained enough experience to be able to improve its performance and thus become more autonomous.

Dealing with real-world uncertainty or robustness is another important characteristic of an intelligent system. Uncertainty arises from many sources, among which are nonlinear behaviour, time-varying behaviour (e.g. degradation over time) and interaction with noisy environments. All of these features are present in human behaviour and therefore are important in the context of machines that interact and co-operate with people in specific domains. It is clear that humans are not as affected by noise in sensory data as are present-day computing machines. For example, recognising a familiar face in a crowd is performed almost instantaneously — studies of the human brain have shown that the pre-attentive processing of stimuli is carried out in as few as 70 to 100 ms. In order to achieve such performance, it is likely that face recognition and many other perceptual and motor tasks (e.g. riding a bicycle) are performed by humans without reliance on explicit rules and without even the subject knowing an explicit description of the computation underlying such processes. We look, see, pay attention and then recognise without directly using rules. On the other hand, humans may accomplish 'higher' cognitive functions such as reasoning, decision-making, planning and communication by using explicit representations. An intelligent machine should therefore be able to combine signal level (sub-symbolic) processing with more abstract reasoning (which could be symbolic) in a coherent manner to result in overall intelligent behaviour.

Learning/adaptability is the third important feature of an intelligent system. Learning can be viewed as change in the behaviour of a system based on the interaction between the system and its environment, and on the observations by a system of its own behaviour. From a dynamical systems point of view, learning is the rate of change of an analytic function describing the system's behaviour [4]. The analytic function is a mapping from the input to the output space, so it could be defined using a collection of symbolic rules, or just be some real-valued function.

There are many possible views on how the features discussed above can be jointly realised in an intelligent system. In the next section, we propose a specific engineering view of intelligent systems which paves the way for a more specific discussion in subsequent sections on how the features contributing towards intelligent behaviour in human/machine interaction systems can be combined.

2. Intelligent systems — an engineering view

Early attempts at building systems with comparable cognitive capability to the human brain, even in limited domains, have not been as successful as predicted. The problems of knowledge representation, commonsense knowledge and reasoning, real-time problem solving, vision, and speech and language processing have not as yet yielded to classical AI techniques. More research is needed and is being carried out mainly by academic departments in the areas of pattern recognition, logical representation, search, inference, learning from experience, planning, ontology and epistemology. Despite the difficulties, AI has been successfully applied in some areas such as expert systems, classification, planning, scheduling, game playing and robotics.

Initially, the AI community aimed to build artefacts that could compete with human level intelligence on various tasks. An alternative view for building intelligent systems is the tool metaphor [5] in which the human is seen as a skilled craftsperson with the system trying to provide the tools that fit the capabilities of the user and the task to be performed [6]. In this view, the focus is on systems that can augment the cognitive capabilities of humans, i.e. a human/machine symbiosis or human-centred intelligent system (HCIS). HCISs are different in many respects from traditional AI systems in that they do not compete with, or try to match, human level intelligence. Rather, they assist humans in performing cognitive tasks such that the combination of the human and the system has a greater cognitive capacity than each on its own. This is a familiar engineering approach of tool building, i.e. just as cars, trains and planes have increased the speed of our physical movement, telescopes, microscopes and night vision cameras have increased our visual capabilities, and satellites have improved our weather prediction capabilities, we are aiming for systems that can help us with our cognitive tasks such as decision making, reasoning, planning, etc. The focus in this approach is to build human assistants that are not identical to humans but complement them to perform tasks like memorising large amounts of information, performing sophisticated planning and scheduling quickly, learning from humans and automating repetitive tasks. The main difference, however, between the mechanical or optical tools mentioned above and HCISs is that the former are passive tools, i.e. they do not exhibit any degree of autonomy or proactivity, while these are essential features of an HCIS.

This approach to building intelligent systems is attractive to practitioners and engineers because its goals are more achievable and it lends itself to an incremental development methodology. However, HCISs are very complex due to the inherent uncertainty present in human/machine interaction, such as the vagueness of natural language, the unpredictable nature of interactions, the lack of good models for human thinking processes and the subjectivity of decision-making processes. Flexibility is the key to the success of these systems. Particularly important are flexible human/machine interaction, flexible information processing and flexible information storage and retrieval.

3. Conceptual architecture for HCISs

In this section, we propose a generic architecture for HCISs (Fig 2). It should be noted that this architecture draws from intelligent agent architectures such as those proposed in Russell and Norvig [7], and Jacobsen [8], with the notable difference that the agent environment has been divided into the human user and the external environment (everything but the user). This has been done so that we can concentrate on the specific requirements of intelligent systems that interact with humans. A second point to note is that this is a general architecture and some of the com-

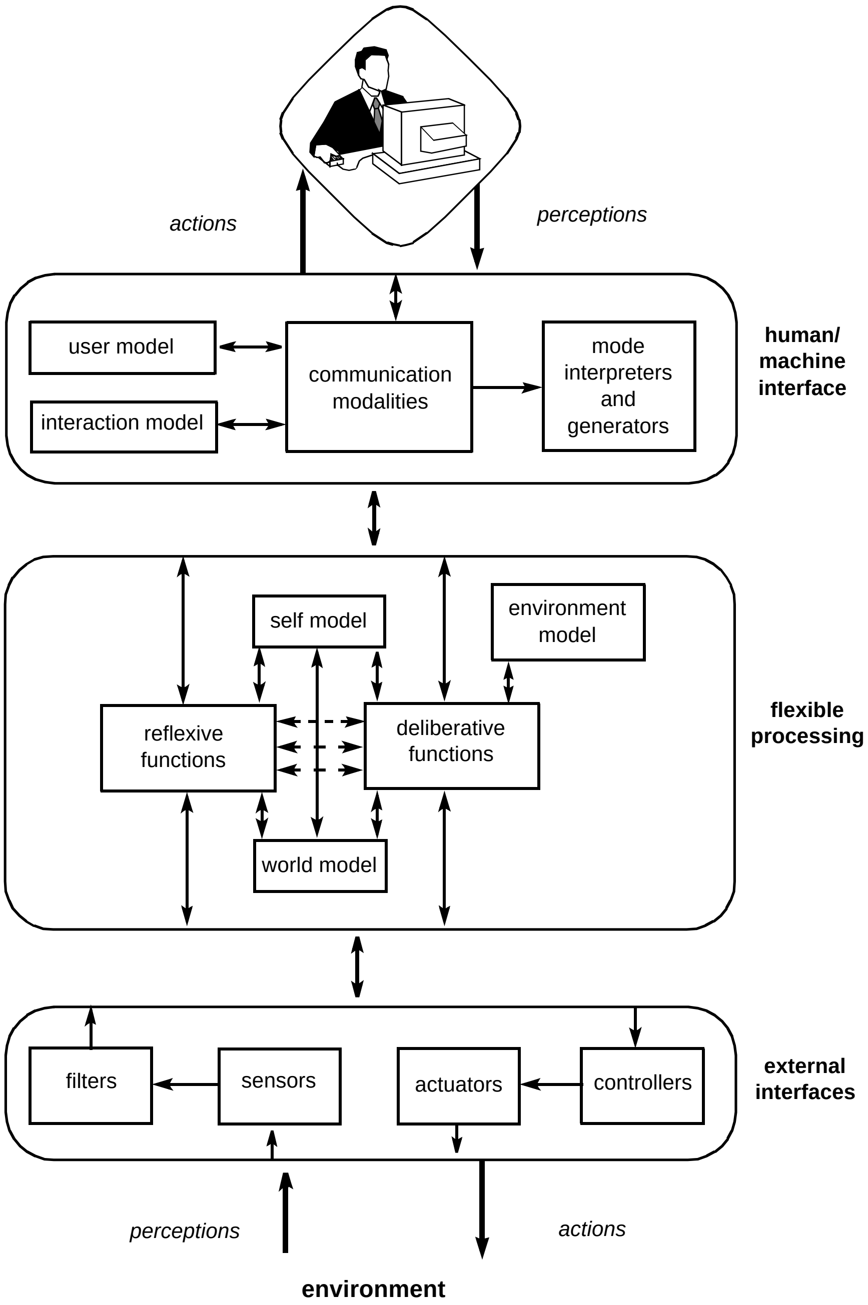

Fig 2 A conceptual architecture for human-centred intelligent systems.

ponents shown in Fig 2 may not be present in some applications. What makes a system intelligent is the knowledge contained within it and the way it manipulates the knowledge to perform goal-oriented tasks. This knowledge can be either implicit in the algorithms and architecture of the system, or explicit in a declarative form. In either case, the knowledge can be learnt from examples or entered by humans. A system is limited in its learning capability by the learning techniques used and its knowledge representation scheme. While open-ended learning is an interesting but unsolved problem, sub-symbolic methods lend themselves more readily to learning from example, but their black-box approach makes it difficult to know what they have learnt. Symbolic approaches are transparent but are inefficient and slow in learning. Hybrid approaches have been developed to learn interpretable rules from examples. Many such techniques result in a large number of rules, which defeats the purpose because humans can neither check them nor learn from them. Clustering and compression techniques then become very important.

An essential component of an HCIS is the human/ machine interface. For a computer to naturally and effectively communicate with humans, it must possess some of the human communication capabilities — e.g. vision, hearing and speech — and must understand natural language. One key point is that the combination of these capabilities enables humans to communicate efficiently. Limiting the communication by blocking some of these modalities (e.g. using the keyboard alone) clearly reduces the amount of information that can be conveyed. This lost information, although redundant in some communications, can be vital in disambiguating vague instructions such as ‘send this to him’. Multi-modal interfaces can therefore go beyond the limitations of present-day interfaces by incorporating general and application-specific knowledge to fuse the information from different devices. In this way, the interface becomes flexible enough to be able to understand human interactions often best expressed using different modalities (e.g. two utterances accompanied by different facial expressions could have vastly differing meanings). So the main reason for using multi-modality in a system is to provide a richer set of channels through which the user and computer can communicate.

Within the human/machine interface, there are major opportunities for exploitation of intelligent technologies. Natural language processing, speech recognition, vision and text-to-speech are a few active areas of research which are central to building HCISs. The synergy of different modalities is one of the major challenges in engineering human interfaces. Fusion of multi-modal information requires low-level learning and temporal reasoning to obtain the temporal relationship between events, and high-level reasoning with application-specific knowledge. Another challenging area of the human interface is to accurately represent and maintain a user model. This involves learning preferences and habits from examples and then representing them in a compact and efficient way. A final important component of the user interface is the interaction model. This usually contains knowledge about the details of interactions such as the syntax, vocabulary, grammar, etc. Often the flexibility of the interface depends on the range of interactions represented within this model.

Both user and interaction models are vital to the degree of autonomy and robustness of HCISs for two reasons:

- in a real operating environment it is not feasible to have access to all the states required in the perception/action sequences — some (hidden) states may have to be derived from the observable states using the models,
- such models maintain learned associations and common action sequences obtained from the history of perception/action sequences and hence facilitate evolvability over time to improve the system’s performance.

Flexible processing provides the coupling between the human and external systems. For a system to have a flexible processing capability it requires:

- knowledge about its own state and abilities (self model),
- knowledge about the environment with which it interacts (environment model),
- knowledge about the world in which it operates (world model),
- mechanisms for deliberative actions such as planning and reasoning,
- mechanisms for reflexive actions such as pattern recognition and reactive decision making.

In the self model, there must be an explicit or implicit way of representing goals and a mechanism to assess the performance of the system in relation to its goals. The world model includes information about the world within a specific domain, such as associations between physical objects or words (e.g. cars and roads). The environment model includes the state of the environment and functions that describe the way the states change with time and as a result of actions taken by systems operating in the environment. The self and environment models are necessary to do ‘what-if’ type analysis, while the world model provides background and commonsense knowledge in general. For example, when driving a car on the motorway the self model represents my driving capabilities, such as my comfortable driving speed, and goals, such as going to the supermarket, while the environment model represents the cars around my car and how they behave, such as overtaking, indicating, and the world model tells me that cars do not fly, they have human drivers, and crashes should be avoided at all costs.

There are many techniques that can be employed to engineer the components of the HCIS architecture shown in Fig 2. In the next three sections, we concentrate on one of the most promising approaches which has had a significant commercial impact on the industrial application of intelligent techniques, known as soft computing.

4. Soft computing — definition and rationale

One of many promising approaches for developing hybrid intelligent systems is soft computing (SC). The term soft computing was coined by Lotfi Zadeh, the inventor of fuzzy set theory. Zadeh defines SC as a partnership between fuzzy logic (FL), neuro-computing (NC), probabilistic reasoning (PR), with the latter subsuming genetic algorithms (GA), chaotic systems, belief networks and parts of learning theory [9].

According to Zadeh, the theory of fuzzy sets represents an attempt to construct a conceptual framework for a systematic treatment of vagueness and uncertainty due to fuzziness in both quantitative and qualitative ways [10]. Zadeh advocates that SC has the means to extend what conventional AI has achieved in the past 40 years, and that a prerequisite to building an intelligent machine is a model of human cognitive capability [11, 12].

The philosophical argument for SC is stated elegantly by Mamdani [13]. It is obvious that humans deal with uncertain and imprecise information every day and are remarkably consistent in processing such information. This is the primary aim of SC, to exploit the tolerance of imprecision, uncertainty and 'partial truth' associated with almost every aspect of real-world problems.

Another concise definition for SC is a term that describes a collection of techniques capable of dealing with imprecise, uncertain or vague information. What brings soft computing techniques together, according to Bonissone [14], is:

> "...their departure from classical reasoning and modelling approaches that are usually based on Boolean logic, crisp classification and deterministic search."

Exact techniques based on symbolic AI work in ideal problems where the system to be modelled or controlled is described by complete and precise information.

Table 1 highlights the main reasons for adopting a hybrid approach within the SC framework [15]. There are five categories used to compare the advantages and dis-advantages of the various approaches (learning and optimisation refer to sub-symbolic learning and optimisation). The techniques have been assessed on the basis of whether learning and optimisation are implicit features or have to be built in. Knowledge extraction refers to symbolic knowledge extraction as defined in conventional AI systems. Real-time operation is linked with implementation issues, i.e. whether each method lends itself readily to hardware implementation or not. Knowledge representation is either symbolic or numeric. Note that the entries in the table assume a simplistic categorisation of types of system, whereas in reality, a particular system could satisfy a property to greater or lesser degree.

Two observations can be made from Table 1. Firstly, there is a good case for combining fuzzy/probabilistic systems, artificial neural networks (ANNs) and evolutionary systems for building intelligent systems because the methods can complement one another. Secondly, such a combination can enhance the capabilities of conventional AI systems. Some proponents of SC [16] use this as a strong argument for developing new ways of producing hybrid systems to address the shortcomings of symbolic AI.

It has been argued [11] that the success of SC (as indicated by an explosion of applications in the present decade) is due to its emphasis on computational intelligence (CI) which is for the most part numeric rather than symbolic. Bezdek [17] defines CI as the first step of achieving biological, or human level, intelligence (BI), and it is purely based on numerical computation using sensory signals. AI lies somewhere between CI and BI and can be achieved by extending CI with symbolic representation and manipulation of non-numeric data (see Table 2). Fuzzy models are particularly suitable for a smooth transition between CI and AI because they can accommodate both numeric and symbolic information in a common framework.

Table 2 The ABC of intelligence [17].

complexity

complexity

Biological Human knowledge + sensory inputs	BNN	BI
Artificial Symbolic + numeric + sensor data	ANN	AI
Computational Numeric	CNN	CI

Bezdek presents the case for combining symbolic and sub-symbolic techniques by introducing a distinction between computational neural networks (CNNs) and ANNs.

Table 1 Relative merits of fuzzy/probabilistic, ANN, evolutionary and conventional AI systems.

	Learning	Knowledge extraction	Real-time operation	Knowledge representation	Optimisation
Fuzzy/probabilistic reasoning	No	Yes	Yes	Symbolic/numeric	No
ANN	Yes	No	Yes	Numeric	No
Evolutionary systems	Yes	No	No	Numeric	Yes
Conventional AI systems	No	Yes	No	Symbolic/numeric	No

He argues that ANNs result from the combination of CNNs, which are purely based on numerical processing of sensory data, and some knowledge usually in the form of non-numeric rules [17].

5. The role of soft computing in HCISs

We have already stated the importance of models in HCISs (e.g. user model, environment model and world model). Real-world problems are typically ill-defined, difficult to model, and with large-scale solution spaces [14]. In these cases precise models are impractical, too expensive or non-existent. Therefore there is a need for approximate modelling techniques capable of handling imperfect and sometimes conflicting information. Soft computing offers a framework for integrating such techniques (see Fig 3).

Models of any system and its environment rely heavily on external measurements. These measurements originate from sensory information. SC enables us to preprocess sensory information, reason using symbolic rules and learn directly from observations. It bridges the gap between low-level sub-symbolic knowledge representation necessary at the measurement and control level, and high-level symbolic representation necessary for deliberative functions.

At the measurement level, soft computing techniques are inherently robust. For example, fuzzy production systems interpolate the output space to provide a continuous surface and are therefore tolerant to small variations of the input quantities.

Learning and adaptation are important features of SC techniques. Fuzzy and probabilistic techniques can be used as the basic mechanism for implementing the reflexive and deliberative functions of Fig 2. Approximate reasoning is required in the user model where the information from the user is incomplete, uncertain or vague. Evolutionary computing can be used to learn both structure and parameters of the models in the HCIS architecture. Neuro-fuzzy techniques take advantage of efficient neural network learning algorithms to learn fuzzy rules and adjust their parameters [18]. In the next section we will give specific examples of the application of SC in HCISs.

6. Application of soft computing in HCISs

Fuzzy set theory has been used to deal with the vagueness of natural language in human/machine communication [19]. Farreny and Prade, for example, introduced possibility and necessity measures to query knowledge bases, so that objects that possibly satisfy a request and those that more or less certainly satisfy a query can be easily retrieved. A typical query might be: ‘Find the students who are very good in Maths and who have approximately the same level in Physics.’

In other applications, a request is transformed progressively using synonymous requests until it can be directly answered from the knowledge base [20]. In this process, the words or expressions which have similar meaning according to a fuzzy measure are used to find an answer more or less appropriate to the initial query. Similarly, in a dialogue with a robot, commands such as pick up the small box on the dark seat can be interpreted by soft computing techniques. Wahlster has simulated a language generation module that uses fuzzy descriptions [21]. It is important to note that soft computing techniques only deal with the vagueness aspect of natural language, and there are many other non-trivial problems in natural language systems.

There are applications of fuzzy techniques to speech-recognition systems where a phonetic fuzzy recogniser has been used to enhance the recognition capabilities of an acoustic-phonetic type speech-recognition system [22].

A soft computing framework for general system modelling in finance applications has been developed by Lee and Smith [23]. In this approach, the authors rely on a hybrid architecture employing both genetic algorithms and neuro-fuzzy techniques to search both structure and parameter space. They conclude that effective modelling techniques must facilitate integration of prior knowledge but retain sufficient flexibility to refine, augment or even discard prior knowledge, thus highlighting the importance of the complementary nature of individual soft computing techniques and their interaction with each other within a particular application domain.

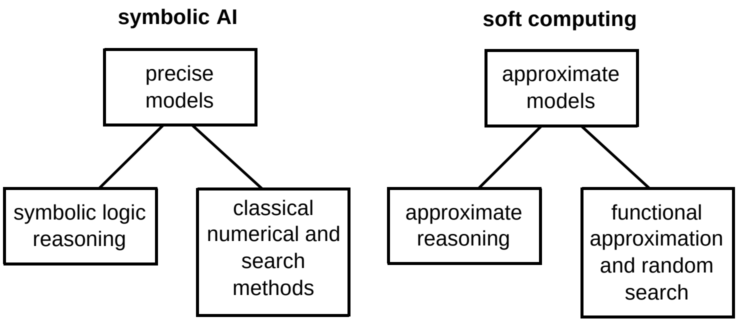

Fig 3 Models in symbolic AI and soft computing paradigms [14].

Mitaim and Kosko [24] have proposed a neural fuzzy system that can learn a user profile in the form of fuzzy if-then rules to be used within a smart software agent. Their work addresses two issues:

- how soft computing can be used to learn what one likes and dislikes,
- how soft computing can be used to search databases on one's behalf.

They have shown how neural-fuzzy function approximators can both help learn a user preference map and choose preferred search objects. Active research issues are accelerating the learning process, and reducing the number of required user interactions.

Soft computing has been used in intelligent retrieval systems to model, using fuzzy linguistic terms, the relevance of index terms to documents using rules such as if the frequency of term *t* in document *d* is high and the frequency of *t* in the archive is low, then *t* is very significant in *d* [25]. The retrieval activity for vague queries can be seen as a decision-making process in which the alternatives are vaguely characterised. The decision process ranks the alternatives according to their possibility and necessity in satisfying the criteria [26]. A framework based on fuzzy pattern matching has been successfully applied to flexible queries in uncertain or imprecise databases [27].

Finally, soft computing techniques have been used at the human interface to enhance the recognition capabilities of computers. Kitzaki and Onisawa have developed a human/computer interaction model based on facial expressions, situations and emotions [28].

7. Future directions — How can we make existing systems more intelligent?

As mentioned earlier, adaptability and robustness are two essential features of an HCIS. Here, we define flexibility as a combination of these two features:

flexibility = adaptability + robustness

To qualify this, let us look at these features more closely. A system is said to be robust if it can tolerate changes to its environment and the way it interacts with its environment without having to change. A system is adaptable if it learns from past experience by changing itself in order to better cope with future situations. It is clear that these two features can be in conflict. One of the key distinguishing factors of intelligent behaviour is knowing how to mix adaptability and robustness. Within the context of HCISs, we should specify the conditions under which learning should occur and rely on robustness under all other possible conditions. For example, consider the problem of travelling between two geographical locations, A and B. Let us assume that there are three known routes: X, Y and Z, but only two are known to Bob (X and Y). If X is the usual way Bob takes between A and B, but he finds X to be blocked by road works, he will take Y because he knows Y is a valid route. So for changes in conditions of X and Y, the X or Y strategy works and is said to be robust. However, if both X and Y are blocked then Bob searches for other routes and learns Z. In this case, the strategy has been adapted to include Z. Now X, Y and Z form a new more robust strategy than X or Y. This simple example shows that adaptation and robustness can be combined to provide flexibility to deal with new or unforeseen conditions.

7.1 Flexible human interfaces

Advances in the fields of vision, speech recognition, text-to-speech and graphical interfaces would greatly enhance the flexibility and the range of capabilities of existing user interfaces. Intelligent interfaces can fuse the multi-modal information and use their task-specific knowledge to overcome the limitations of single modalities. Our experiments have shown that speech recognition accuracy improves dramatically by simplifying words used during dialogue, e.g. instead of 'give me more information about object 02134', we can say 'tell me more about this' where 'this' refers to a graphical representation of object 02134 on the screen at which the user is looking. From the human point of view, flexibility of a system can be measured in terms of naturalness of interactions. There are situations in which pointing or gestures are more natural ways of specifying a query than natural language, for example in direct manipulation of objects or selecting an object out of many represented on the screen. Handling the inherent flexibility within natural language is also an important issue in an HCIS. Humans express the same meaning in many different ways, e.g. the playback message on an answering machine. Given a system that performs relatively few functions (e.g. an automated telephone answering system), increased flexibility of the system means a many-to-few mapping of possible queries to system functions. When dealing with more complicated interaction this mapping becomes non-trivial as the number of possible queries increases exponentially. Natural language understanding systems try to use the meaning of a sentence to simplify this task. The so-called 'do what I mean' (DWIM) interfaces are one way to address this issue. Another more challenging problem with natural language is the ambiguity, i.e. multiple possible interpretations. Resolving such ambiguities by simultaneous parallel evaluation or by engaging in clarification dialogue is another way to make existing systems more flexible.

7.2 Flexible processing

Flexibility at the interface enables the HCIS to obtain a user query or goal. Flexible processing transforms a user's goal into a sequence of sub-goals or actions that the system can satisfy. Intelligent systems should exhibit adaptive goal-oriented behaviour [29]. Systems with such a capability use plans to convert the problem from 'what' to 'how'. This problem has been extensively studied within AI (e.g. systems such as report generators and database query by example methods use template-based plans). More generally, intelligent systems must be able to create for themselves a set of goals given a set of known operations and methods [29]. What is interesting is the behaviour of the system when the goal cannot be satisfied. For example, consider the problem of contacting a person, e.g. Bob, by phone. Here the goal (i.e. talking to Bob by phone) can be divided into the following sub-goals:

- finding Bob's whereabouts (e.g. office),
- finding his telephone number,
- dialling the number,
- talking to Bob.

What if Bob is not at his desk? Depending on the urgency of the call people could take a number of actions — hang up and try later, leave a message and hang up, keep trying until Bob returns to his desk, or try to call Bob's colleague who sits near Bob to find out if Bob is about, etc. The above plans may not satisfy the original goal, but they are all possible ways to contact Bob. Therefore flexibility can be viewed as the ability of the system to revise its actions to achieve the goal or a goal similar to the original one. What should the HCIS do if no goal within an acceptable set of goals can be achieved? This could occur if the initial goal is ambiguous or if the goals fail. User interaction is necessary in such situations. The HCIS should have the capability to acquire, represent and utilise new knowledge based on observing the user in order to learn from experience. Learning is essential to keep an accurate up-to-date model of both the system and the world. Deliberative functions such as planning and reflexive functions such as pattern recognition can be learnt through observations.

8. Conclusions

Intelligent behaviour is hard to characterise. From an engineering point of view, we are interested in specific features and definitions that would allow us to incrementally build systems that are more intelligent than their predecessors. In this paper we have identified autonomy, robustness and adaptability as three important features of existing systems that can be improved. Furthermore, given an overall architecture for a particular class of intelligent system called an HCIS, we have argued that a collection of techniques known as soft computing (SC), the emphasis of which is on approximate solutions, is an ideal candidate for incorporating flexibility into HCISs. We have described a number of examples where SC has been successfully applied to HCISs.

Soft computing is a very promising but young technology in comparison to computer science and AI (which themselves are very young compared to physics, chemistry, mathematics, etc). It has enjoyed a relatively successful life mainly due to its application in the area of control systems. HCISs present us with very different types of problem, mainly studied within the realms of AI. From an engineering point of view, SC and AI techniques can benefit from both closer co-operation and integration. We believe this is the only way in which the true potential of soft computing techniques, within the context of HCIS, can be realised.

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Wayne Wobcke was awarded a BSc with joint Honours in Mathematics and Computer Science (1984) and a research MSc (1986) in Computer Science from the University of Queensland, and a PhD in Computer Science (1989) from the University of Essex. He has 9 years experience as a lecturer in the Basser Department of Computer Science, University of Sydney, where he taught undergraduate courses on artificial intelligence, logic programming and compilers, and an advanced course on natural language processing. His main research is on the application of logic in artificial intelligence to formalise problems such as conditional reasoning, belief revision, plan recognition, and more recently, agent-based computing. He has recently organised a number of workshops concerned with the foundations of intelligent agent systems.

His current research aims to develop a theory of plan and intention revision within agent-based systems, and to implement an agent architecture instantiating the theory. Part of the motivation for the work is to understand the relationship between soft computing and more standard models of computation, and to investigate the use of hybrid techniques for building artificial intelligence systems.